

Ultraviolet astronomy

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A [GALEX](#) image of the [spiral galaxy Messier 81](#) in [ultraviolet](#) light. Credit:GALEX/[NASA/JPL-Caltech](#).

Ultraviolet astronomy is the observation of [electromagnetic radiation](#) at [ultraviolet](#) wavelengths between approximately 10 and 320 [nanometres](#); shorter wavelengths—higher energy photons—are studied by [X-ray astronomy](#) and [gamma-ray astronomy](#).^[1] Ultraviolet light is not visible to the [human eye](#).^[2] Most of the light at these wavelengths is absorbed by the Earth's atmosphere, so observations at these wavelengths must be performed from the upper atmosphere or from space.^[1]

Ultraviolet [line spectrum](#) measurements ([spectroscopy](#)) are used to discern the chemical composition, densities, and temperatures of the [interstellar medium](#), and the temperature and composition of hot young stars. UV observations can also provide essential information about the [evolution of galaxies](#). They can be used to discern the presence of a hot [white dwarf](#) or [main sequence](#) companion in orbit around a cooler star.^{[3][4]}

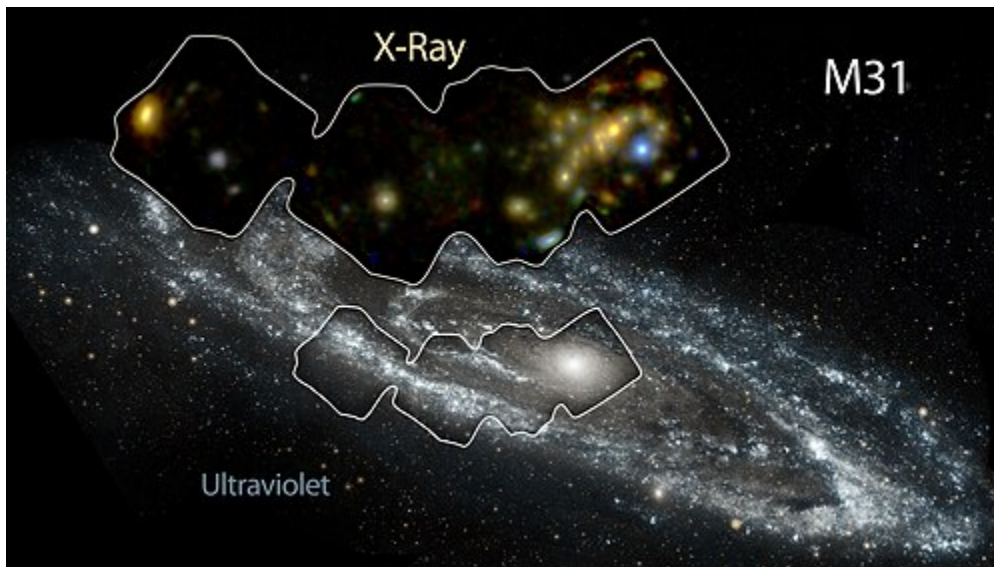
The ultraviolet [universe](#) looks quite different from the familiar [stars](#) and [galaxies](#) seen in [visible light](#). Most stars are actually relatively cool objects emitting much of their electromagnetic radiation in the visible or near-[infrared](#) part of the spectrum. Ultraviolet radiation is the signature of hotter objects, typically in the early and late stages of their [evolution](#). In the Earth's sky seen in ultraviolet light, most stars would fade in prominence. Some very young massive stars and some very old stars and galaxies, growing hotter and producing higher-energy radiation near their birth or death, would be visible. Clouds of gas and dust would block the vision in

many directions along the [Milky Way](#).

Space-based solar observatories such as [SDO](#) and [SOHO](#) use ultraviolet telescopes (called [AIA](#) and [EIT](#), respectively) to view activity on the Sun and its [corona](#). Weather satellites such as the [GOES-R](#) series also carry [telescopes](#) for observing the Sun in ultraviolet.

The [Hubble Space Telescope](#) and [FUSE](#) have been the most recent major [space telescopes](#) to view the near and far UV [spectrum](#) of the sky, though other UV instruments have flown on smaller observatories such as [GALEX](#), as well as [sounding rockets](#) and the [Space Shuttle](#).

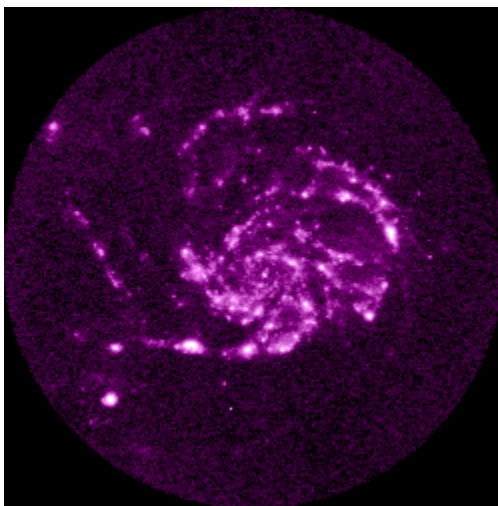
Pioneers in ultraviolet astronomy include [George Robert Carruthers](#), [Robert Wilson](#), and [Charles Stuart Bowyer](#).



[Andromeda Galaxy](#) - in [high-energy X-ray](#) and ultraviolet light (released 5 January 2016).

Ultraviolet space telescopes

[\[edit\]](#)



Astro 2 UIT captures [M101](#) with ultraviolet shown in purple

 - [Far Ultraviolet Camera/Spectrograph](#) on [Apollo 16](#) (April 1972)

-  +  - [TD-1A](#) (135-286 nm; 1972–1974)
-  - [Orbiting Astronomical Observatory](#) (#2:1968-73. #3:1972-1981)
-  - [Orion 1 and Orion 2 Space Observatories](#) (#1: 200-380 nm, 1971; #2: 200-300 nm, 1973)
-  +  - [Astronomical Netherlands Satellite](#) (150-330 nm, 1974–1976)
-  +  - [International Ultraviolet Explorer](#) (115-320 nm, 1978–1996)
-  - [Astron-1](#) (150-350 nm, 1983–1989)
-  - Glazar 1 and 2 on [Mir](#) (in [Kvant-1](#), 1987–2001)
-  - [FAUST](#) (140-180 nm, in ATLAS-1 [Spacelab](#) aboard [STS-45](#) mission, March 1992)^[5]
-  - [EUVE](#) (7-76 nm, 1992–2001)
-  - [FUSE](#) (90.5-119.5 nm, 1999–2007)
-  +  - [Extreme ultraviolet Imaging Telescope](#) (on [SOHO](#) imaging Sun at 17.1, 19.5, 28.4, and 30.4 nm)
-  +  - [Hubble Space Telescope](#) (various 115-800 nm, 1990-1997-) ([STIS](#) 115–1030 nm, 1997–) ([WFC3](#) 200-1700 nm, 2009–)
-  - [Swift Gamma-Ray Burst Mission](#) (170–650 nm, 2004-)
-  - [Hopkins Ultraviolet Telescope](#) (flew in 1990 and 1995)
-  - [ROSAT](#) XUV^[6] (17-210eV) (30-6 nm, 1990–1999)
-  - [Far Ultraviolet Spectroscopic Explorer](#) (90.5-119.5 nm, 1999–2007)
-  - [Galaxy Evolution Explorer](#) (135–280 nm, 2003–2012)
-  - [Hisaki](#) (130-530 nm, 2013 - 2023)
-  - [Lunar-based ultraviolet telescope \(LUT\)](#) (on [Chang'e 3](#) lunar lander, 245-340 nm, 2013 -)
-  - [Astrosat](#) (130-530 nm, 2015 -)
-  - [Colorado Ultraviolet Transit Experiment](#) - (255-330 nm spectrograph, 2021-)
-  - [PROBA-3](#) (CUTE) - (530-588 nm coronagraph, 2024-)
-  - Public Telescope (PST)^[7] (100-180 nm, Proposed 2015, EU funded study)
-  - Viewpoint-1 SpaceFab.US (200-950 nm, Launch planned 2022)^[8]
-  - [Carruthers Geocorona Observatory](#)

See also [List of ultraviolet space telescopes](#)

Ultraviolet instruments on planetary spacecraft

[\[edit\]](#)

[Markarian galaxies](#) – Galaxy with a nucleus emitting exceptionally large amounts of ultraviolet

[Pea galaxy](#) – Possible type of luminous blue compact galaxy

[^] [Jump up to: ^a ^b](#) A. N. Cox, ed. (2000). *Allen's Astrophysical Quantities*. New York: Springer-Verlag. ISBN 0-387-98746-0.

[^] ["Ultraviolet Light"](#). Archived from [the original](#) on 2017-02-13. Retrieved 2017-02-12.

[^] Reimers, D. (July 1984). "Discovery of a white dwarf companion of the "hybrid" K giant HD 81817". *Astronomy and Astrophysics*. **136**: L5–L6. [Bibcode:1984A&A...136L...5R](#).

[^] Ortiz, Roberto; Guerrero, Martín A. (September 2016). ["Ultraviolet emission from main-sequence companions of AGB stars"](#). *Monthly Notices of the Royal Astronomical Society*. **461** (3): 3036–3046. [arXiv:1606.09086](#). [Bibcode:2016MNRAS.461.3036O](#). [doi:10.1093/mnras/stw1547](#).

[^] Lampton, Michael; Sasseen, Timothy P.; Wu, Xiaoyi; Bowyer, Stuart (1993). "A study of the impact of the space shuttle environment on faint far-UV geophysical and astronomical phenomena". *Geophysical Research Letters*. **20** (6): 539–542. [Bibcode:1993GeoRL..20..539L](#). [doi:10.1029/93GL00093](#).

[^] ["R. Staubert, H. Brunner,1 H.-C. Kreysing - The German ROSAT XUV Data Center and a ROSAT XUV Pointed Phase Source Catalogue \(1996\)"](#). Archived from [the original](#) on 2015-01-20. Retrieved 2012-01-20.

[^] [Ein privates Weltraumteleskope für Amateure und Profis](#) Spektrum DE. June 2015

[^] ["Space Telescopes"](#).

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